

RIGOROUS TECHNICAL DOSSIER // NEUTRAL CROSS-EVALUATION DOCUMENT

Smartphone Intelligent Dead Reckoning (IDR) with GNSS Fusion

Independent Technical Specification, Algorithmic Implementation, Empirical Evidence, and Verification Audit

ACTIVE REPOSITORY	HEAD COMMIT	EVALUATION TARGET	EVALUATION SEED
Recursive-Minds/manas-phase3	4d2c974 (Sep 16, 2026)	Drift < 10% Travelled Distance	541098 (Single-Seed Protocol)

DOCUMENT SECTIONS	
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1. PROJECT OVERVIEW

1.1 Operational Context & Problem Formulation

This project implements a complete, sensor-agnostic **Smartphone Intelligent Dead Reckoning (IDR) and GNSS Fusion Engine** designed for continuous, high-accuracy vehicle navigation during complete Global Navigation Satellite System (GNSS) signal outages. Such outages occur routinely in real-world driving through highway tunnels, underground parking structures, urban street canyons flanked by high-rise towers (severe multipath and satellite obscuration), elevated multi-tier flyovers, and dense roadside forest canopies.

In consumer vehicle navigation, standard inertial double-integration of unconstrained smartphone accelerometer data suffers from rapid quadratic and cubic position drift within 3 to 5 seconds due to sensor bias instability, low-frequency 1/f noise, and thermal drift. The Smart India Hackathon (SIH) benchmark specification defines an operational target requirement:

- **Drift Target:** Final horizontal position drift must remain strictly under **10% of total distance travelled** throughout complete GNSS blackout periods (e.g. under 5 meters of drift over 50 meters of travel, or under 100 meters over 1 kilometer of travel).
- **Hardware Constraints:** Purely software-based edge solution executing on consumer-grade mobile devices using internal MEMS sensors (10 Hz 3-axis accelerometer, 10 Hz 3-axis gyroscope, magnetometer) without physical access to vehicle Controller Area Network (CAN) bus odometry, wheel ticks, or external high-precision IMU units.

1.2 System Inputs and Outputs

- **Inputs:**
 - Continuous raw smartphone inertial data: 3-axis acceleration `[a_x, a_y, a_z]` (m/s²) and 3-axis angular rates `[w_x, w_y, w_z]` (rad/s) sampled at 10 Hz.
 - Pre-blackout GNSS fixes (WGS84 latitude, longitude, altitude, ground speed, course over ground bearing) arriving at ~0.11 Hz (9.0-second intervals in raw logs, with causal 1.0 Hz pre-blackout historical interpolation).

- Vector road network representation: Ordered road segments with geographic start/end coordinates, segment bearing, length, road classification (Highway, Arterial, Urban), and AASHTO curvature design standards.
- **Outputs:**
 - Local 2D East-North-Up (ENU) coordinates `[p_east, p_north]` and geodetic coordinates (latitude, longitude).
 - Vehicle forward speed (m/s) and forward speed uncertainty (variance).
 - Vehicle chassis yaw heading (degrees relative to True North).
 - Map-matched corridor coordinates snapped orthogonally or topologically to the active road network.
 - Real-time confidence metrics, error covariances, and handoff reconciliation states.

1.3 Implementation Team Context & Neutrality Statement

This repository (`Recursive-Minds/manas-phase3`) represents one of several independent, equal-effort implementations of the same problem built by different teammates working from distinct architectural hypotheses, model designs, and fusion strategies. It was developed to enable an honest, transparent, and rigorous technical comparison across approaches without prejudice or bias toward any individual implementation. No claim of system superiority, ranking, or comparative verdict is made or implied in this dossier; all findings herein reflect solely the verifiable code, configurations, logs, and empirical data present in this workspace.

2. ARCHITECTURE

2.1 Final End-to-End Pipeline Architecture

The final production architecture is implemented in the modular `sih/` package and orchestrated by `sih/engine/dead_reckoning_engine.py`. The data flow connects nine distinct stages behind strict interface contracts:

```
[Raw Smartphone IMU @ 10 Hz]
|
▼
[Stage 1: Mount Auto-Calibrator (sih/calibration/mount.py)]
- Gravity extraction during rest (accel norm ~ 9.81 m/s^2, gyro norm < 0.05 rad/s)
- Centripetal yaw axis identification: cross(a_lat, v_fwd)
- Real-time SO(3) coordinate rotation: Body Frame → Vehicle Chassis Frame
|
▼
[Stage 2: AI Forward Velocity Estimator (sih/models/moe_fusion.py)]
- 12-channel input tensor (calibrated kinematics + spectral vibration powers)
- Dual-Expert Bayesian Mixture of Experts:
  * Expert 1 (ResNet1DSpeedEstimator): 20-sample (2.0s) window, 4 residual blocks
  * Expert 2 (TCNAttentionVelocityModel): 60-sample (6.0s) window, dilated conv + self-attention
  * Bayesian Gating Network: Softmax expert weights + heteroscedastic log-variance
|
▼
[Stage 3: Causal Speed Smoother (sih/fusion/speed_smoother.py)]
- Slew-rate limiting: +3.5 m/s^2 acceleration cap, -5.0 m/s^2 braking deceleration cap
- 1st-order causal EMA low-pass filter: tau = 0.25s (alpha = 0.286 at 10 Hz)
|
▼
[Stage 4: Road Kinematics Governor (sih/map/governor.py)]
- AASHTO / IRC civil highway design curvature speed caps:
  v_max = min( sqrt(a_lat_max / kappa), a_lat_max / |omega_z| )
- Enforces a_lat_max = 1.2 m/s^2 (Highway) or 3.5 m/s^2 (Arterial/Urban)
|
▼
[Stage 5: 15-State Error-State Kalman Filter (sih/fusion/es_ekf.py)]
- Error states: [delta_p(3), delta_v(3), delta_theta(3), b_a(3), b_g(3)]
- Inertial strapdown propagation @ 10 Hz
```

```

- Measurement Updates:
  * AI forward velocity update with dynamic model uncertainty
  * Closed-loop Non-Holonomic Constraints (NHC):  $v_{\text{lateral}}^{\text{body}} = 0$ ,  $v_{\text{vertical}}^{\text{body}} = 0$ 
  * Zero-Velocity Updates (ZUPT): triggers when  $v_{\text{fwd}} < 0.2$  m/s and accel variance is low
    |
    ▼
[Stage 6: Pre-Blackout Heading & Scale Seeder (sih/engine/dead_reckoning_engine.py)]
- Synthesizes causal 1.0 Hz historical pre-blackout GNSS stream (zero future lookahead)
- Seeds initial heading from moving GPS velocity vector ( $v > 2.5$  m/s) + forward gyro integration
- Computes pre-blackout speed scale factor:  $\text{mean}(v_{\text{GPS}}) / \text{mean}(v_{\text{AI}})$  bounded to  $[0.85, 1.38]$ 
    |
    ▼
[Stage 7: Topological HMM Map Matcher (sih/map/matcher.py)]
- Spatial candidate search within 35-meter radius
- Domain-differentiated snapping:
  * Highway & Arterial: Strictly perpendicular lateral correction (preserves along-track DR)
  * Urban Grid: Corner projection with topological successor transition across 90-degree turns
- Multi-hypothesis scoring: exponential distance-heading likelihood with turn-intent gating
    |
    ▼
[Stage 8: Phase 6 Seamless GNSS-INS Handoff FSM (sih/handoff/manager.py)]
- 5-State Finite State Machine (GNSS_HEALTHY, BLACKOUT_DETECTED, INS_DEAD_RECKONING, etc.)
- C2 quintic polynomial spline trajectory smoothing across a 3.0-second reconciliation window
    |
    ▼
[Stage 9: Edge Mobile Causal Streaming Runtime (sih/mobile/causal_stream.py)]
- 10 Hz ring-buffer sliding window execution
- PyTorch TorchScript compilation: models/exported/moe_velocity_model.torchscript.pt (2.66 MB)

```

2.2 Component Specifications & Dimensional Contracts

Subsystem / Module	Source File	Input Dimensions	Output Dimensions	Key Physical / Numerical Parameters
Generic Data Loader	sih/data/loader.py	Raw CSV rows	IMUSample , GNSSSample	Decoupled Android/sensor contracts; handles 10 Hz to 200 Hz rates.
Mount Calibrator	sih/calibration/mount.py	IMUSample (6-axis)	CalibratedSample (6-axis + norm)	min_samples=30 ; gravity_magnitude=9.81 m/s^2 ; SO(3) matrix.
Dual-Expert MoE	sih/models/moe_fusion.py	(B, 12, 20) & (B, 12, 60)	v_pred (B, 1) , log_var (B, 1)	ResNet 64 ch (4 blocks); TCN 32 ch (dilations 1,2,4,8; 4 heads); 1.97M params.
Speed Smoother	sih/fusion/speed_smoother.py	Scalar v_raw (float)	Scalar v_smooth (float)	a_max = +3.5 m/s^2 , a_min = -5.0 m/s^2 , tau = 0.25s (alpha=0.286).
15-State ES-EKF	sih/fusion/es_ekf.py	IMU tick + VelocityEstimate	FusedPosition (ENU pos, vel, quat)	15 error states; R_nhc = diag([0.1, 0.1]) ; turn_thresh = 2.5 deg/s .
Road Governor	sih/map/governor.py	Speed, curvature kappa , yaw rate	Governed speed v_gov	AASHTO comfort limits: a_lat_max = 1.2 m/s^2 (Hwy), 3.5 m/s^2 (Urb).

Subsystem / Module	Source File	Input Dimensions	Output Dimensions	Key Physical / Numerical Parameters
HMM Map Matcher	<code>sih/map/matcher.py</code>	<code>FusedPosition</code> + <code>RoadNetwork</code>	<code>MatchedPosition</code>	Search radius 35m; lateral snap gate; turn-intent heading unlock (>20 deg).
Handoff Manager	<code>sih/handoff/manager.py</code>	Dead-reckoned pos + Reacquired GNSS	Smoothed output trajectory	C2 quintic spline transition; window = 3.0s; <code>sigma_gate</code> = 3.0.

2.3 Alternative Architectures Investigated, Abandoned, or Changed

Traceable from git commit history (`git log`), archived experiment notes, and `CODEBASE_DIFFERENCES_AUDIT_REPORT.md`, several earlier architectural configurations were implemented and subsequently replaced:

1. Pure 6-Axis Double Integration (Phase 1 Baseline):

- *Implementation:* Naive strapdown integration of phone accelerometers without external velocity constraints (`benchmarks/run_naive_baseline.py`).
- *Outcome:* Rapid divergence with drift exceeding 60% P90 (median 22.89% across 40 scenarios). Unconstrained accelerometer bias error integrated twice produced cubic position growth in < 10 seconds.
- *Reason for abandonment:* Confirmed that dead reckoning with consumer MEMS IMUs is physically impossible without independent forward velocity measurements and closed-loop constraints.

2. Kinematic ES-EKF without AI Velocity Estimation (Phase 2):

- *Implementation:* 15-state ES-EKF with NHC and ZUPT, but relying on integrated forward accelerometer values rather than an AI speed estimator (`benchmarks/run_phase2_es_ekf.py`).
- *Outcome:* Better than double integration, but drift still exceeded 20% to 35% on highway stretches due to road vibration noise rectifying into false forward acceleration.
- *Reason for abandonment:* Forward accelerometer integration cannot distinguish chassis tilt/grade from true acceleration.

3. Single-Expert TCN-Attention Model (`best_velocity_model.pt`):

- *Implementation:* Dilated 1D TCN with multi-head attention trained on 8 calibrated IMU channels with SmoothL1 loss (checkpoint size 4.78 MB).
- *Outcome:* Severe high-speed underprediction. On highway test sequences, the predicted speed scale ratio was $\text{sum}(v_pred) / \text{sum}(v_GT) = 0.787$ (a 21.3% systemic speed deficit; validation RMSE was 4.23 m/s).
- *Reason for abandonment:* Huber/SmoothL1 loss compressed gradients on large errors occurring at high speeds (> 20 m/s), forcing the network toward lower average predictions. Replaced by the Dual-Expert MoE architecture and the Rule 8 balanced loss.

4. 25-Second Pre-Blackout Dynamic Scale Estimator:

- *Implementation:* In early revisions of `benchmarks/run_final_benchmark.py`, a 25.0-second lookback window was used, requiring at least `N >= 4` moving GPS fixes ($v > 2.5$ m/s) to update the scale factor.
- *Outcome:* Because phone GPS in the IO-VNBD dataset arrived at ~9.0-second intervals, a 25-second window contained at most 2 fixes. The `N >= 4` check failed in 100% of scenarios, leaving the scale factor frozen at 1.000 (mathematically dead code).
- *Reason for change:* Identified in code audit; replaced by pre-blackout 1.0 Hz GNSS stream synthesis that interpolates valid historical fixes over 15 to 25 seconds, guaranteeing sufficient sample support.

5. Comparison with Sister Fork (`Recursive_Minds-SIH26168`):

- *Divergence:* The sister fork abandoned explicit mount calibration and the heavy MoE model entirely, developing **CausalSpeedNet** (155k parameters, 1.92 MB) using 14 rotation-invariant features (norm of acceleration, projected

forward acceleration, vibration powers, inclination angles) trained with CAN-bus supervision.

- *Local Choice*: This workspace retained explicit SO(3) coordinate calibration and the Dual-Expert MoE model (7.52 MB checkpoint), while adopting CAN-bus time alignment offsets and causal speed smoothing.

2.4 Stated Technical Rationale for Final Architecture Selection

Per project documentation in `SYSTEM_STATE_AND_ARCHITECTURE.md` and `sih/models/moe_fusion.py`:

- **Complementary Frequency Modeling**: A single temporal network cannot simultaneously capture high-frequency pavement vibration cues (which correlate with tire roll speed across 2 to 10 Hz) and macro-scale vehicle momentum profiles (acceleration ramps across 5 to 10 seconds). The Dual-Expert MoE assigns high-frequency transients to the ResNet-1D (2.0s window) and long-range trends to the Dilated TCN (6.0s window), with gating weights dynamically adjusting between highway cruising and stop-and-go conditions.
- **Newtonian Closed-Loop Grounding**: Rather than predicting 2D trajectory positions end-to-end via deep learning (which fails to generalize across unseen road geometries), AI is restricted strictly to scalar forward speed. Position and heading integration are delegated to the 15-state ES-EKF governed by immutable kinematic constraints (NHC: zero lateral/vertical body velocity).
- **Civil Engineering Curvature Bounds**: Empirical observations demonstrated that open-loop gyro bias accumulation during prolonged outages caused artificial centrifugal drift. Incorporating AASHTO / IRC road design standards ($v_{\max} = \sqrt{a_{\text{lat_max}} / \kappa}$) imposes a physical speed ceiling through turns.

3. TRAINING METHODOLOGY

3.1 Mathematical Formulation of Implemented Loss Functions

The training objective is implemented in `sih/models/losses.py`. Two primary composite formulations exist in the repository:

A. High-Speed Balanced Velocity Loss (`balanced_velocity_loss`)

Implemented for single-expert TCN training to satisfy Rule 8 (scale-balanced velocity loss):

```
Loss = L_smooth_l1 + w_scale * L_scale + 0.5 * L_high + w_dyn * L_dyn + 0.1 * L_var

Where:
1. L_smooth_l1 = SmoothL1(y_pred, y_true, beta = 1.0)
2. L_scale     = ( (sum(y_pred) / (sum(y_true) + 1e-4)) - 1.0 )^2
3. L_high      = mean( (y_pred[y_true > 8.0] - y_true[y_true > 8.0])^2 )
4. L_dyn       = clamp( (Var(y_true) - Var(y_pred)) / max(Var(y_true), 1.0), min = 0.0, max = 10.0 )
5. L_var       = SmoothL1( exp(clamp(log_var, -4.0, 4.0)), (y_pred.detach() - y_true)^2, beta = 1.0 )

Hyperparameters as implemented:
w_scale = 2.0, w_dyn = 1.0, beta = 1.0, v_high_threshold = 8.0 m/s (28.8 km/h).
```

B. Dual-Expert Bayesian MoE Loss (`phase55_balanced_loss`)

Multi-objective composite loss used for training the champion Mixture-of-Experts checkpoint:

```
Loss_total = L_primary + 2.0 * L_scale + 0.5 * L_high + 0.10 * L_var + w_dyn * L_dyn [+ 0.05 * L_cent] [+ w_cls * L

Where:
1. L_primary   = SmoothL1(v_fused, v_gt, beta = 1.0)
                + 0.5 * SmoothL1(v_res, v_gt, beta = 1.0)
                + 0.5 * SmoothL1(v_tcn, v_gt, beta = 1.0)
2. L_scale     = clamp( ( (mean(v_fused) / clamp(mean(v_gt), min = 0.5)) - 1.0 )^2, max = 25.0 )
3. L_high      = clamp( mean( (v_fused[v_gt > 8.0] - v_gt[v_gt > 8.0])^2 ), max = 50.0 )
```

```
4. L_var      = SmoothL1(var_fused, (v_fused.detach() - v_gt)^2, beta = 1.0)
               + 0.5 * SmoothL1(var_res, (v_res.detach() - v_gt)^2, beta = 1.0)
               + 0.5 * SmoothL1(var_tcn, (v_tcn.detach() - v_gt)^2, beta = 1.0)
5. L_dyn      = clamp( (Var(v_gt) - Var(v_fused)) / max(Var(v_gt), 1.0), min = 0.0, max = 10.0 )
6. L_cent     = mean( (a_lateral - v_fused * omega_yaw)^2 )   [active if centripetal signals present]
7. L_cls      = CrossEntropy(class_logits, motion_labels)     [active if motion classification head enabled]
```

Hyperparameters as implemented:
w_dyn = 2.0, w_cls = 0.2, beta = 1.0.

3.2 Hyperparameter Evolution & Historical Rescaling

- **High-Speed Loss Domination:** Initially, standard Mean Squared Error (MSE) led to training instability on vibration noise, while SmoothL1 (beta=1.0) caused the network to underpredict high speeds by 21%. The addition of `L_scale` with weight `w_scale = 2.0` penalized systemic bias without destabilizing local gradients.
- **Dynamics Variance Alignment (`L_dyn`):** Early iterations produced overly conservative models that collapsed toward the global trip mean velocity (~12 m/s). The asymmetric variance deficit loss (which fires only when predicted variance is lower than ground truth variance) was introduced with weight `w_dyn = 2.0` to force the model to follow acceleration transients.

3.3 Training Configuration & Optimizer Specifications

Parameter	Specification in Code (<code>train_velocity_model.py</code>)	Hardware & Concurrency Safeguards
Optimizer	<code>torch.optim.AdamW</code>	Decoupled weight decay regularization.
Learning Rate	Initial: <code>1.5e-3</code> ; eta_min: <code>1e-5</code>	<code>CosineAnnealingLR</code> over 25 epochs.
Weight Decay	<code>1e-4</code>	L2 parameter penalty on linear and convolutional weights.
Batch Size	Default: 64 (safe for 8GB VRAM)	Dynamic CUDA OOM recovery: auto-reduces to 32 or 16.
Epochs	25 epochs	Saves best checkpoint based on: <code>RMSE + 5.0 * scale - 1.0 </code> .
Precision	Mixed Precision FP16 via <code>torch.amp.autocast('cuda')</code>	<code>GradScaler</code> with unscale norm clipping at <code>max_norm = 5.0</code> .
Concurrency	<code>num_workers = 0</code> , <code>pin_memory = False</code>	Prevents Windows multiprocessing page-lock crashes.
Data Augmentation	3D SO(3) random rotation matrices applied to IMU samples	Enforces cradle mount angle invariance during training.

3.4 Random Seed Traceability & Sensitivity Audit

CRITICAL RIGOR FINDING: SINGLE-SEED EVALUATION PROTOCOL

- Model Training Scripts:** Examination of `train_velocity_model.py`, `train_5fold_cross_validation.py`, and `scripts/train_can_moe.py` reveals that **no explicit random seed was set** (no `torch.manual_seed` or `np.random.seed` call exists in the training loops). Checkpoints represent single unseeded stochastic runs.
- Benchmark Evaluation Scripts:** In `benchmarks/run_final_benchmark.py` and `scripts/compare_checkpoints.py`, a **single fixed random seed (541098)** is hardcoded as default. All headline benchmark numbers reported in this project stem exclusively from seed `541098`.
- Seed Sensitivity (Cross-Audit Evidence):** As documented in `CODEBASE_DIFFERENCES_AUDIT_REPORT.md`, when the sister fork evaluated 5 independent seeds (`541098`, `42`, `123456`, `9999`, `77777`) across 200 scenarios, median drift varied significantly: from **8.83%** (Seed 42) to **14.05%** (Seed 541098), yielding a pooled median of **12.73%**. In the current local workspace, multi-seed evaluation was never executed; multi-seed statistical variance is therefore **untested and unconfirmed** in this repository.

4. DATA AND SPLITS

4.1 Datasets and Ingestion Profiles

The repository utilizes real-world vehicle drives from the **Indian On-Road Vehicle Navigation Benchmark Dataset (IO-VNBD)**, stored locally in `data/raw/ionvnbdr_trips/`. The dataset features smartphone IMU recordings along with phone GNSS fixes and ground truth vehicle CAN-bus wheel speeds:

Trip ID	Operational Environment	IMU Samples (10 Hz)	GNSS Fixes (~0.11 Hz)	Duration	10 Hz CAN Ground Truth
<code>S-M.csv</code>	Highway Cruising (Multi-lane, high speed)	105,974	1,178	~2.9 hours	Available (<code>V-M.csv</code> , offset +2.3s)
<code>S-S1.csv</code>	Urban City Grid (Stop-and-go, 90° turns)	51,746	575	~1.4 hours	Available (<code>V-S1.csv</code> , offset +0.2s)
<code>S-S2.csv</code>	Arterial Corridors (Medium speed, chicanes)	93,876	1,043	~2.6 hours	Available (<code>V-S2.csv</code> , offset +8.6s)
<code>S-S3a.csv</code>	Mixed Urban / Arterial Test Drive	~120,000	~1,330	~3.3 hours	Available (<code>V-S3a.csv</code> , offset -6.9s)
<code>S-S4.csv</code>	Arterial Suburban Test Drive	~80,000	~890	~2.2 hours	Not Available in IO-VNBD (Synchronous)

4.2 Partitioning Scheme & Embargo Buffers

Partitioning is managed by `sih/data/split.py` (`compute_trip_partition`):

- Development Sequences (`S-M` , `S-S1` , `S-S2`):** Evaluated using a 3-way temporal split:
 - Part 1: Training (60%):** Indices `0` to `0.60 * N`. Used for model weight optimization.
 - Embargo Buffer 1 (15 seconds / 150 samples):** Completely discarded boundary purge to eliminate autocorrelation.
 - Part 2: Validation (20%):** Indices `0.60 * N + 150` to `0.80 * N`. Used for model checkpoint selection.
 - Embargo Buffer 2 (15 seconds / 150 samples):** Completely discarded boundary purge.
 - Part 3: Benchmark (20%):** Indices `0.80 * N + 150` to `N`. Used for scenarios #01 through #20.
- Completely Unseen Test Sequences (`S-S3a` , `S-S4`):** 100% held-out test drives. Zero samples were ever exposed to training or validation. Used for scenarios #21 through #40.

4.3 Explicit Data Leakage Audit: Mechanisms Found, Fixed, and Unfixed

A rigorous inspection of the code, data handling, and network extraction revealed three distinct mechanisms affecting data independence:

LEAKAGE MECHANISM 1: WITHIN-TRIP TEMPORAL PARTITIONING (PARTIALLY ADDRESSED)

Mechanism: For `S-M`, `S-S1`, and `S-S2`, training (Part 1) and benchmarking (Part 3) originate from the *same continuous trip recording*. Although 15-second embargo buffers prevent temporal bleeding of overlapping window features, the phone cradle mount orientation, vehicle suspension harmonics, sensor calibration bias, and pavement texture are identical between Part 1 and Part 3.

Status: Unfixed for sequences `S-M`, `S-S1`, `S-S2`. However, sequences `S-S3a` and `S-S4` (which constitute 50% of the 40-scenario benchmark) were held out entirely, verifying that the system can execute on completely unseen vehicle drives.

LEAKAGE MECHANISM 2: SELF-REFERENTIAL ROAD NETWORK CONSTRUCTION (CRITICAL FINDING)

Mechanism: In `benchmarks/run_final_benchmark.py` line 198, the road network for each trip is constructed via `build_road_network(trip, prefix)`, which calls `build_road_network_from_trip(trip)` in `sih/map/network.py`. This function constructs the road network directly from the **valid GNSS trajectory of the vehicle itself**, decimated to 10-meter steps.

Implication: The road network passed to the map matcher during the blackout is literally a 1D corridor of the car's own true path. There are no false branching streets, parallel frontage roads, or wrong turns in the network graph. Map-matching drift is effectively lateral snapping to the ground truth track. The system's ability to disambiguate complex multi-lane road forks or open-world OpenStreetMap networks is therefore not tested by the 40-scenario benchmark.

LEAKAGE MECHANISM 3: 9-SECOND GPS STAIR-STEPPING & LAG ILLUSION (DIAGNOSED & FIXED)

Mechanism: Smartphone GNSS fixes arrive every ~9.0 seconds. Between fixes, the velocity column in raw CSVs repeats identical values for 90 rows. Naive linear interpolation created artificial 5-second lags and 17-meter sagitta cut-offs on highway curves.

Fix: Corrected by integrating 10 Hz vehicle CAN-bus wheel speed logs (`V-*.csv`) and computing cross-correlation time offsets (`S-S1: +0.2s`, `S-S2: +8.6s`, `S-M: +2.3s`, `S-S3a: -6.9s`), aligning ground truth with true vehicle motion.

4.4 Cross-Team Evaluation Comparability Notice

Warning regarding direct score comparison: A future peer review cannot directly compare the raw drift percentages in this report against a teammate's project unless both teams evaluated on the exact same scenario time windows, used identical ground-truth speed references (CAN vs interpolated phone GPS), and tested against equivalent road network topologies (self-generated polyline vs external OSM graphml). Cross-evaluation must assess methodological rigor, physical plausibility, and architectural soundness rather than raw numerical scores alone.

5. EVALUATION AND RESULTS BY SCENARIO

5.1 Headline Benchmark Performance Summary

Evaluated on 40 standardized GNSS blackout scenarios across 5 distinct trips on seed `541098` (verified directly against `artifacts/phase4_multi_trip_benchmark_results.csv`):

Evaluation Metric	Pure 6-Axis IMU Baseline	Phase 4 Map-Matched Pipeline	Benchmark Target	Compliance Status
Overall Median Drift (%)	22.89%	11.33%	< 10.0%	11.33% (NEAR TARGET)

Evaluation Metric	Pure 6-Axis IMU Baseline	Phase 4 Map-Matched Pipeline	Benchmark Target	Compliance Status
P90 (Worst Decile) Drift (%)	60.25%	34.50%	< 35.0%	PASSED
Tier 1 Pass Rate (< 10% Drift)	12.5% (5 / 40)	47.5% (19 / 40)	> 50.0%	47.5% (NEAR TARGET)
High Reliability Rate (<= 30% Drift)	65.0% (26 / 40)	85.0% (34 / 40)	> 85.0%	PASSED
Initial Heading Seeding Error	28.40° (unobservable)	Median 0.001° / Mean 6.53°	< 2.0° median	PASSED

5.2 Operational Multi-Tier Breakdown

Operational Regime	Speed & Distance Range	Outage Duration	Multi-Trip Benchmark Result	Official Benchmark Threshold	Status
Tier 1: Traffic Crawl	< 20 km/h / < 200m	30s – 60s	49.1m Median Position Error	< 10m absolute error	NEAR TARGET
Tier 2: City Maneuvers	20 – 50 km/h / 200m – 500m	30s – 60s	7.04% Median Drift	< 15% of distance travelled	PASSED
Tier 3: Highway Cruising	> 50 km/h / 500m – 1.2 km	60s – 75s	11.90% Median Drift	< 100m over 1km (< 10%)	NEAR TARGET

5.3 Domain & Sequence Performance Breakdown

Driving Domain	Scenarios	Pure 6-Axis Median	Phase 4 Map Median	Tier 1 Pass Rate (<10%)	Sub-30% Pass Rate	Compliance Status
Arterial	16	24.18%	9.88%	8/16 (50.0%)	13/16 (81.2%)	PASSED
Highway	8	21.69%	7.16%	7/8 (87.5%)	8/8 (100.0%)	PASSED
Mixed	10	24.11%	14.12%	3/10 (30.0%)	8/10 (80.0%)	NEAR TARGET
Urban	6	38.37%	24.32%	1/6 (16.7%)	5/6 (83.3%)	NEAR TARGET

5.4 Duration Breakdown & Non-Monotonic Trend Analysis

Blackout Duration	Scenarios	Pure 6-Axis Median	Phase 4 Map Median	Tier 1 Pass Rate (<10%)	Sub-30% Pass Rate
30s	12	28.68%	14.12%	5.0/12 (41.7%)	11.0/12 (91.7%)
45s	12	20.66%	9.55%	7.0/12 (58.3%)	9.0/12 (75.0%)
60s	8	27.36%	10.39%	4.0/8 (50.0%)	6.0/8 (75.0%)
75s	8	22.89%	11.38%	3.0/8 (37.5%)	8.0/8 (100.0%)

OPEN CAVEAT: NON-MONOTONIC DURATION DRIFT DISTRIBUTION

Notice that the median drift for **30-second outages (14.12%)** is noticeably higher than for **45-second outages (9.55%)** and **60-second outages (10.39%)**. In a theoretical dead-reckoning system, drift percentage is expected to grow monotonically with duration. In this benchmark, shorter outages correspond to shorter absolute distances travelled (often 100m to 250m). At low distances, fixed initial orthogonal snap offsets (3m to 10m) constitute a disproportionately high percentage of total travel distance. This is noted as an unresolved characteristic of ratio-based drift metrics on short runs rather than evidence of superior long-duration integration.

5.5 Complete 40-Scenario Granular Evaluation Table

Detailed results for all 40 scenarios evaluated on seed `541098` from `artifacts/phase4_multi_trip_benchmark_results.csv`:

ID	Trip Sequence	Domain	Duration	Distance	Pure Drift	Map Drift	Gain	Map Error	Hdg Seed Err	Classification
#01	S-M (Highway)	Highway	30s	296.6m	41.66%	1.77%	+39.89%	5.25m	15.81°	TIER 1 (<10%)
#02	S-M (Highway)	Highway	60s	719.1m	21.39%	11.90%	+9.49%	85.61m	0.00°	SUB-30%
#03	S-M (Highway)	Highway	60s	1064.0m	40.21%	8.88%	+31.33%	94.43m	14.48°	TIER 1 (<10%)
#04	S-M (Highway)	Highway	45s	948.1m	21.98%	9.86%	+12.12%	93.53m	0.00°	TIER 1 (<10%)
#05	S-M (Highway)	Highway	30s	456.5m	10.60%	4.10%	+6.51%	18.71m	0.00°	TIER 1 (<10%)
#06	S-M (Highway)	Highway	75s	619.7m	22.94%	6.60%	+16.34%	40.91m	0.00°	TIER 1 (<10%)
#07	S-M (Highway)	Highway	75s	1045.1m	21.19%	7.71%	+13.47%	80.61m	3.50°	TIER 1 (<10%)
#08	S-M (Highway)	Highway	45s	527.8m	18.81%	6.43%	+12.38%	33.94m	0.00°	TIER 1 (<10%)
#09	S-S2 (Arterial)	Arterial	45s	383.5m	23.05%	34.43% DEGRADED	-11.38%	132.05m	0.00°	>30%
#10	S-S2 (Arterial)	Arterial	75s	663.8m	39.34%	19.79%	+19.55%	131.36m	0.00°	SUB-30%
#11	S-S2 (Arterial)	Arterial	30s	248.1m	18.54%	19.87% DEGRADED	-1.33%	49.29m	0.00°	SUB-30%
#12	S-S2 (Arterial)	Arterial	45s	338.5m	25.32%	4.61%	+20.71%	15.59m	28.27°	TIER 1 (<10%)
#13	S-S2 (Arterial)	Arterial	30s	198.4m	14.51%	9.02%	+5.49%	17.89m	11.61°	TIER 1 (<10%)
#14	S-S2 (Arterial)	Arterial	60s	795.2m	33.33%	1.91%	+31.43%	15.16m	8.65°	TIER 1 (<10%)
#15	S-S1 (Urban)	Urban	45s	367.8m	19.33%	27.84% DEGRADED	-8.51%	102.40m	23.54°	SUB-30%
#16	S-S1 (Urban)	Urban	60s	519.2m	57.42%	49.11%	+8.31%	254.95m	7.78°	>30%
#17	S-S1 (Urban)	Urban	75s	613.2m	5.49%	24.41% DEGRADED	-18.93%	149.70m	0.00°	SUB-30%
#18	S-S1 (Urban)	Urban	45s	459.7m	6.52%	24.23% DEGRADED	-17.70%	111.36m	1.95°	SUB-30%
#19	S-S1 (Urban)	Urban	30s	233.8m	105.67%	19.53%	+86.14%	45.66m	0.00°	SUB-30%

ID	Trip Sequence	Domain	Duration	Distance	Pure Drift	Map Drift	Gain	Map Error	Hdg Seed Err	Classification
#20	S-S1 (Urban)	Urban	30s	124.7m	96.89%	0.00%	+96.89%	0.00m	2.24°	TIER 1 (<10%)
#21	S-S3a (Mixed)	Mixed	30s	225.1m	26.85%	26.81%	+0.04%	60.34m	0.00°	SUB-30%
#22	S-S3a (Mixed)	Mixed	60s	660.7m	8.19%	6.12%	+2.07%	40.46m	0.00°	TIER 1 (<10%)
#23	S-S3a (Mixed)	Mixed	30s	195.4m	13.87%	13.71%	+0.16%	26.80m	0.00°	SUB-30%
#24	S-S3a (Mixed)	Mixed	75s	1229.7m	22.84%	12.02%	+10.82%	147.78m	0.00°	SUB-30%
#25	S-S3a (Mixed)	Mixed	75s	1716.6m	25.38%	20.60%	+4.78%	353.62m	0.00°	SUB-30%
#26	S-S3a (Mixed)	Mixed	30s	604.2m	31.13%	14.52%	+16.61%	87.74m	0.00°	SUB-30%
#27	S-S3a (Mixed)	Mixed	45s	511.9m	16.18%	9.23%	+6.95%	47.28m	0.00°	TIER 1 (<10%)
#28	S-S3a (Mixed)	Mixed	45s	207.1m	51.15%	39.66%	+11.49%	82.13m	17.58°	>30%
#29	S-S3a (Mixed)	Mixed	60s	200.1m	85.78%	30.42%	+55.37%	60.85m	39.40°	>30%
#30	S-S3a (Mixed)	Mixed	45s	249.8m	10.47%	6.34%	+4.14%	15.83m	0.00°	TIER 1 (<10%)
#31	S-S4 (Arterial)	Arterial	75s	585.8m	26.47%	10.75%	+15.72%	62.97m	0.00°	SUB-30%
#32	S-S4 (Arterial)	Arterial	30s	345.0m	4.30%	0.43%	+3.87%	1.47m	0.00°	TIER 1 (<10%)
#33	S-S4 (Arterial)	Arterial	75s	505.6m	13.26%	7.64%	+5.62%	38.63m	5.29°	TIER 1 (<10%)
#34	S-S4 (Arterial)	Arterial	30s	234.9m	30.52%	27.28%	+3.23%	64.10m	33.06°	SUB-30%
#35	S-S4 (Arterial)	Arterial	45s	370.0m	17.26%	8.55%	+8.72%	31.62m	7.34°	TIER 1 (<10%)
#36	S-S4 (Arterial)	Arterial	30s	125.1m	43.25%	39.26%	+3.99%	49.13m	9.22°	>30%
#37	S-S4 (Arterial)	Arterial	45s	392.7m	118.24%	2.30%	+115.93%	9.04m	0.00°	TIER 1 (<10%)
#38	S-S4 (Arterial)	Arterial	45s	734.2m	36.74%	35.16%	+1.58%	258.17m	1.16°	>30%
#39	S-S4 (Arterial)	Arterial	60s	494.6m	5.64%	1.14%	+4.50%	5.64m	14.82°	TIER 1 (<10%)
#40	S-S4 (Arterial)	Arterial	60s	1087.5m	11.41%	15.26% DEGRADED	-3.85%	165.90m	15.42°	SUB-30%

5.6 Detailed Analysis of Degradation Scenarios

In **6 out of 40 scenarios (15.0%)**, the Map-Matched pipeline produced *higher* endpoint drift than the unconstrained Pure 6-Axis IMU EKF:

Scenario	Trip	Duration	Distance	Pure Drift	Map-Matched Drift	Degradation Delta
#09	S-S2 (Arterial)	45s	383.5m	23.05%	34.43%	+11.38%
#11	S-S2 (Arterial)	30s	248.1m	18.54%	19.87%	+1.33%
#15	S-S1 (Urban)	45s	367.8m	19.33%	27.84%	+8.51%
#17	S-S1 (Urban)	75s	613.2m	5.49%	24.41%	+18.93%
#18	S-S1 (Urban)	45s	459.7m	6.52%	24.23%	+17.70%
#40	S-S4 (Arterial)	60s	1087.5m	11.41%	15.26%	+3.85%

ROOT CAUSE OF MAP-MATCHING DEGRADATIONS

- Scenario #17 (S-S1 Urban, 75s, 613m):** Pure EKF drift was 5.49% (33.7m), while Map-Matched drift grew to 24.41% (149.7m), representing an 18.93% performance regression. Root cause: at severe 90-degree urban street grid turns, the corner-projection heuristic lagged behind actual vehicle rotation, snapping the vehicle onto an adjacent parallel street segment and accumulating along-track error.
- Scenario #18 (S-S1 Urban, 45s, 460m):** Pure EKF drift was 6.52% (30.0m); Map Matching drifted to 24.23% (111.4m), a 17.70% regression caused by false projection near an urban junction.
- Scenario #09 (S-S2 Arterial, 45s, 384m):** Pure EKF drift was 23.05%; Map Matching drifted to 34.43% (11.38% regression).

5.7 Protocol-Defined vs. Dataset-Dependent Metrics

- Protocol-Defined Metrics (Cross-Team Comparable):**
 - Endpoint drift percentage ($\text{error_m} / \text{distance_travelled_m} * 100$).
 - P90 worst-decile drift threshold (< 35%).
 - Multi-tier operational regimes (Tier 1 crawl < 20 km/h, Tier 2 city 20-50 km/h, Tier 3 highway > 50 km/h).
- Dataset-Dependent Metrics (Not Directly Cross-Team Comparable):**
 - Absolute meter errors (e.g. 49.1m crawl error): Depends on specific trip speed profiles, traffic pauses, and stop durations.
 - Initial heading seeding error (mean 6.53°): Highly dependent on pre-blackout vehicle dynamics (moving straight vs executing an intersection turn at blackout entry).
 - Map-matched accuracy gain (+11.56% median): Artificially elevated because the benchmark road network was generated from the vehicle's own GNSS track.

6. VERIFICATION AND SELF-CORRECTION

6.1 Diagnostic Self-Corrections & Bug Fixes Documented in Repository

Issue / Bug Caught	Detection Mechanism	Root Cause	Corrective Action Implemented	Empirical Result
Pre-blackout scale estimator dormant	Code inspection & logging	25s lookback window required ≥ 4 fixes; 9s GPS provided at most 2 fixes.	Synthesized causal 1.0 Hz GNSS stream prior to blackout in <code>dead_reckoning_engine.py</code> .	Activated scale estimator across all scenarios; corrected highway asphalt damping.
High-speed AI velocity scale collapse	Validation curve audit (<code>best_velocity_model.pt</code>)	Huber loss gradient saturation on high speeds caused 21% speed underprediction (scale 0.787).	Engineered Dual-Expert MoE and Rule 8 balanced loss with explicit scale penalty ($w=2.0$).	Scale ratio normalized to 1.067; validation RMSE decreased from 4.23 to 3.28 m/s.
9-second GPS stair-step optical illusion	Visual inspection of diagnostic curves	9s GPS values repeated for 90 samples; linear interpolation created false 5s lag and 17m sagitta dips.	Integrated 10 Hz continuous CAN wheel speed logs (<code>V-*.csv</code>) with sub-second time offsets.	Eliminated artificial sagitta errors; validated true continuous speed tracking.
Junction corner lock-in trapping	Deep diagnosis script (<code>diagnose_four_failures.py</code>)	Gyro turn lag caused map matcher to snap to straight roads before turn completed.	Engineered turn-intent gating disabling premature heading re-anchoring at forks.	Resolved branch lock-in in sharp highway forks and chicanes.
Report claim contradiction in summary text	Repository audit of <code>FINAL_JUDGE_EVALUATION_REPORT.md</code>	Summary text on line 232 stated "Achieved overall median drift $< 10\%$ (11.33%)".	Flagged contradiction: 11.33% is greater than 10.0%, and Tier 1 pass rate was 47.5% (19/40).	Clarified that overall system is NEAR TARGET (11.33%), with Highway (7.16%) and Arterial S-S4 (9.65%) passing.

6.2 Claims in Existing Reports Found Unverified or Partially Verified

1. Live Indian Road OSM Ingestion (Partially Verified):

- *Claim:* README and documentation state that the system operates via live Overpass API queries of Indian road vectors (`sih/map/governor.py`).
- *Verification Status:* The Overpass client and cache code exist and pass offline unit tests (`tests/test_map_ingestion.py`). However, **it is NOT used in the primary 40-scenario benchmark**, which runs entirely on self-generated polylines from the UK IO-VNBD dataset (`build_road_network_from_trip`). Live

OpenStreetMap road ingestion on real Indian roads remains an offline architectural demonstration, not a verified benchmark result.

2. Phase 6 Seamless GNSS-INS Handoff FSM (Partially Verified):

- *Claim:* Documentation describes C2 quintic spline trajectory smoothing during blackout exit and GNSS reacquisition.
- *Verification Status:* Implemented in `sih/handoff/manager.py` and verified on synthetic trajectories in `tests/test_handoff.py`. However, the 40 benchmark scenarios terminate at the end of the blackout duration; **the handoff FSM is not exercised in the 40-scenario benchmark run.**

3. 5-Fold Cross-Validation Generalization (Unverified / Negative Evidence):

- *Status:* Inspection of the 5-fold cross-validation checkpoints in `models/checkpoints/5fold/moe_fold_*.pt` revealed that validation scale ratios ranged from **0.402 to 0.727** with an average validation RMSE of **5.867 m/s**. The model showed significant cross-fold sensitivity, indicating that without pre-blackout scale anchoring, the AI velocity estimator struggles to generalize across different trips.

6.3 Scope of Untested Conditions

- **Physical Android On-Device Runtime:** Latency, frame drops, RAM consumption, and thermal throttling have not been profiled on physical Android mobile hardware; evaluation was performed on an x86_64 host with NVIDIA RTX 4060 GPU and Intel CPU.
- **Multi-Seed Statistical Confidence:** The benchmark has only been run on seed `541098` in this repository.
- **Unconstrained Phone Orientations:** Dynamic in-pocket, hand-held, or vibrating cup-holder phone placements have not been tested; the mount calibrator assumes a rigidly affixed cradle.
- **Multi-Level Interchanges:** Grade separations, stacked flyovers, and underground spiral ramps are unverified due to lack of 3D road vector layers.

7. WHAT WOULD STRENGTHEN THIS SYSTEM

An objective technical audit reveals five concrete areas where this system requires strengthening before it could be considered a production-ready automotive dead-reckoning engine:

1. **Decouple Benchmark from Self-Referential Road Networks:** The primary benchmark must be updated to evaluate against external OpenStreetMap or GIS road vector files (such as Coventry or Indian city networks) rather than polylines extracted from the vehicle's own GNSS path. This will rigorously test multi-hypothesis fork disambiguation and junction turn selection under real branching conditions.
2. **Institute Multi-Seed Pooled Evaluation with Statistical Bounds:** Benchmark reporting should transition from the single seed `541098` to a pooled 5-to-10-seed confirmation protocol with reported standard deviations, 95% bootstrap confidence intervals, and paired Wilcoxon signed-rank tests to guard against seed cherry-picking.
3. **Mitigate Low-Speed Urban Crawl Drift (24.32% Median):** The 24.32% median drift in urban environments (reaching 49.11% in Scenario #16) is the system's primary accuracy bottleneck. Integrating an adaptive engine-idle vibration filter and dynamic ZUPT noise floor calibration will prevent false forward distance accumulation during traffic stops.
4. **Enforce Strict Leave-One-Trip-Out (LOTO) Cross-Validation:** Eliminate within-trip temporal splitting on `S-M`, `S-S1`, and `S-S2`. Re-train models strictly on complete, independent trips to demonstrate genuine vehicle- and driver-independent generalizability.
5. **On-Device Android Execution Profiling:** Measure real-time execution time, battery drain, and memory usage for the exported TorchScript model (`moe_velocity_model.torchscript.pt`) on an actual ARM64 mobile device to prove edge feasibility.

8. REPRODUCIBILITY NOTES

8.1 Execution Environment & Key Dependencies

- **Operating System:** Windows 11 (tested) / Linux x86_64 compatible.
- **Python Runtime:** Python 3.10 to 3.13 (active environment: Python 3.13.2).
- **Core Packages:** `torch >= 2.0.0` (CUDA optional, CPU supported), `numpy >= 1.24.0`, `pandas >= 2.0.0`, `matplotlib >= 3.7.0`, `scipy >= 1.10.0`.

8.2 Primary Entry Scripts & Command-Line Invocations

Workflow	Command Line	Key Outputs
40-Scenario Benchmark	<code>python benchmarks/run_final_benchmark.py --seed 541098</code>	Generates CSV results, drift histograms, 40 trajectory plots, and judge report.
A/B Model Comparison	<code>python scripts/compare_checkpoints.py</code>	Head-to-head comparison between Champion and Candidate checkpoints.
MoE Model Training	<code>python train_velocity_model.py --model moe --epochs 25 --batch_size 64</code>	Trains Dual-Expert MoE on S-S1/S-S2/S-M Part 1; outputs checkpoint to <code>models/checkpoints/</code> .
Single-Expert TCN Training	<code>python train_velocity_model.py --model tcn --epochs 25 --batch_size 64</code>	Trains single TCN model; outputs <code>best_velocity_model.pt</code> .
Mobile Streaming Test	<code>python -m unittest tests/test_mobile_stream.py</code>	Tests 10 Hz ring-buffer causal streaming engine and TorchScript inference.
Full Test Suite	<code>python -m unittest discover -s tests -p "test_*.py"</code>	Executes 41 unit and regression tests across all modules (2.08s run time).

CLOSING NOTE

This document is one of two (or more) independent, equally-effort implementations of the same team project, produced for honest, no-bias comparison; that the purpose of comparing them is to select or synthesize the strongest final system by evaluating architecture soundness, methodological rigor, truthfulness/transparency about limitations, testing thoroughness, and verification quality — not test-set scores alone, since test sets differ across implementations; and that ideas, mechanisms, or design choices from either system may be adopted into the other, or combined, en route to a single final submitted system.